

Biomimetic Case Study

BIOE70012 - Biomimetics

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Week 6 – Group A

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A. Introduction

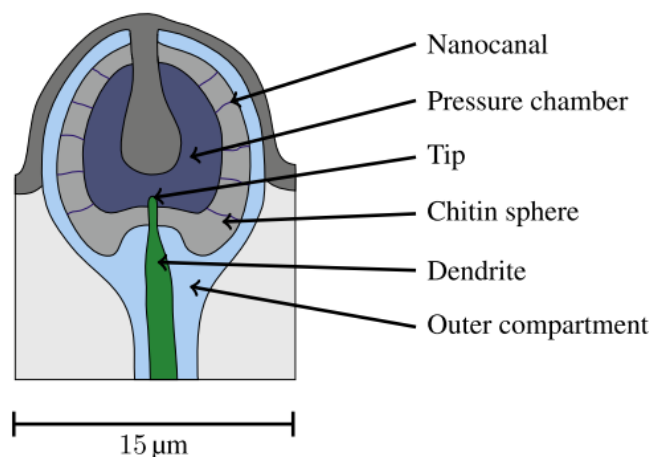
Forest fires can reach temperatures up to 1000 degrees, an environment typically unliveable for most organisms. However, the beetle *melanophila acuminata* has been observed to approach these fires, soon after the fire has gone out, and use the lack of the burnt trees' defence mechanisms to mate and lay their eggs. They have been shown to approach from as far away as 20km away.

The mechanism by which these beetles find and approach these fires is through the detection of infrared radiation, peak activation occurring at a wavelength of 3 micrometers. Forest fires release radiation between 2.2 and 4 micrometers.

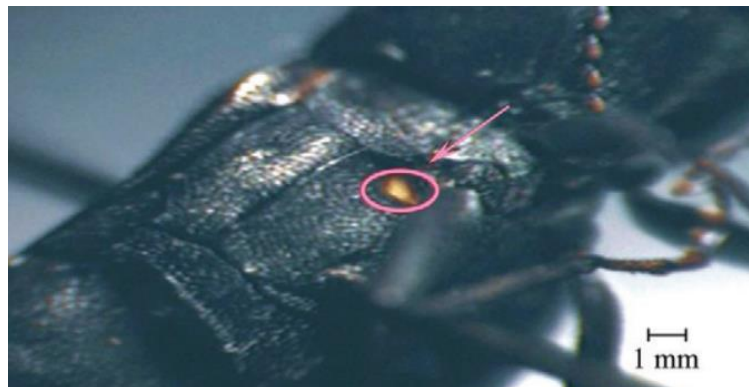
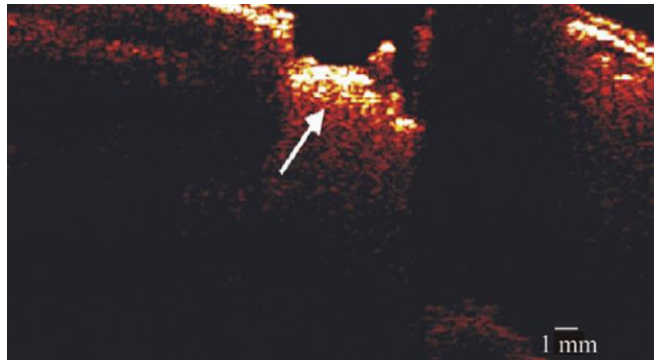
They have specialized sensory pit organs which convert the infrared radiation from the fires to heat, leading to a series of mechanical changes within the receptor, allowing bioelectric nerve transmission.

In more detail, IR radiation is absorbed by chitin fibres and tissue fluid in the receptor, which is converted into heat. Temperature rise and volume expansion of the receptor then takes place. A lamellated layer around the internal structure restrains the expansion, so the hydraulic pressure is increased which deforms the tip of the dendrite of the mechanoreceptor – this is the signal for the bioelectric nerve transmission. The high sensitivity is due to the synergistic effects of the high pressure forming mechanism and the stress concentrated structure of the dendrite tip and compliant membrane.

This highly sensitive and extremely rapid response could form the basis of the development of invaluable sensors for human use. The response occurs within 2 milliseconds and with a threshold value as low as 500 microWatts/cm². The mechanisms involved in this sensory system is worth investigating and replicating for the purpose of sensor development.



B. Methods



As the images show, the organ indicated by the white arrow and pink arrow is beetle's infrared sensory organs, which contains a unique 15 kDa protein that can absorb infrared (IR) radiation at $3\text{ }\mu\text{m}$ and $10\text{ }\mu\text{m}$. These wavelengths match the IR signature of forest fires, making the infrared sensory organs highly selective.

The unique advantage of the infrared sensory organs of beetle is that it does not rely on the change of temperature, and it directly absorbs and detects infrared photons via protein vibration. The photo-effect-based transduction mechanism is also introduced, which is similar to photoreceptors in the eye (infrared light).

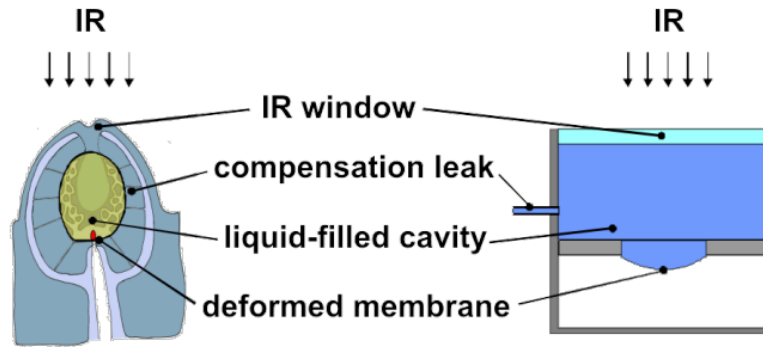
In this new era, the current IR sensors heavily depend on the thermal detection, which means the working principle of these sensors base on the change of temperature. Those sensors can be slow and energy-consuming. Based on the feature of the infrared sensory organs of the beetle, a protein-based detector is better than current IR sensors, which can react immediately to IR photons and potentially being more sensitive, selective, and power-efficient.

Aside from that, the absorption of infrared photons via protein vibration also leads to the change of pressure, reflected by liquid expansion inside infrared sensory organs, which is called the

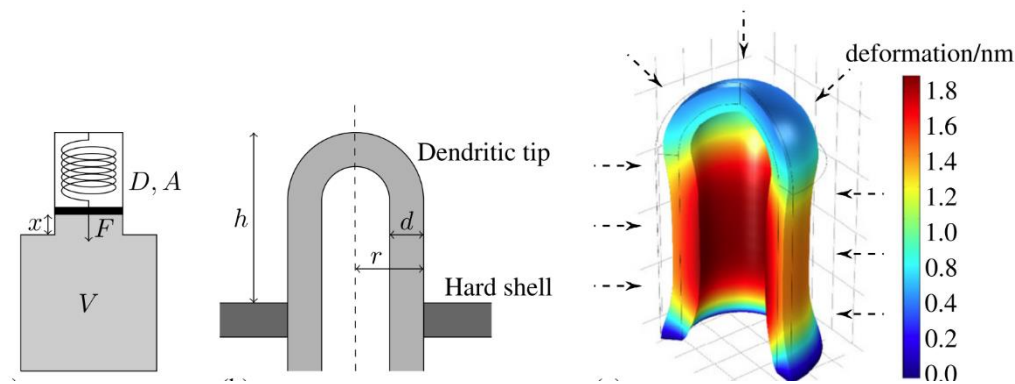
liquid-filled microfluidic core. Therefore, when the thermal expansion occurs, the volume of liquid increases.

The situation of liquid expansion also lead to the pressure buildup, which occurs in a confined space, making even small temperature changes generate significant pressure fluctuations. The dominant pressure change formula for IR sensor is listed below, where ΔP stands for the change of pressure, β stands for the isochoric tension coefficient in the unit of Pa/K, ΔT_{mean} is the change of mean temperature in the cavity, κ is denoted as the isothermal compression coefficient in the unit of Pa^{-1} , V is original volume of the cavity, and ΔV is the change of volume with the change of temperature.

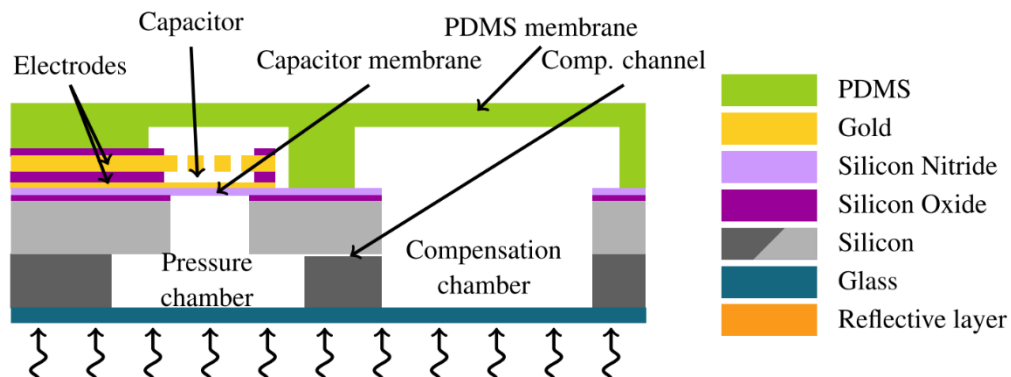
$$\Delta P = \beta \cdot \Delta T_{mean} - \frac{1}{\kappa \cdot V} \cdot \Delta V$$



The physical change of pressure inside the infrared sensory organs converts the energy of change of pressure into the electrical signal, allowing the beetle to detect the IR of thermal expansion. The deformation of the membrane of the liquid-filled microfluidic core stretches ion channels, and these channels open, allowing ions, including (Na^+ , K^+ , Ca^{2+}), to flow into the neuron. The flow of ions depolarizes the neuron, generating an electrical signal. The neuro received the electrical signal fires an action potential (AP), sending the electrical signal to the central nervous system (CNS) of the beetle. The electrical signal encodes the intensity of the IR radiation, allowing the beetle to process the location and strength of a thermal source. For example, the thermal source can be a forest fire.



The image provides a visual representation of the mechanical response of the infrared sensory organs in the beetle when the beetle is exposed to the infrared radiation (IR). It converts the nature working principle in thermal energy detection of the beetle to the artificial uncooled IR sensors (based on the beetle's natural system). The MEMS-based capacitive sensors can be optimized using similar pressure-driven deformation mechanisms.



The biomimetic sensor in the above image replicates the mechanotransduction mechanism found in beetles. The pressure-drive membrane deformation allows the signal delivered to the brain and allows for uncooled infrared detection, eliminating the need for expensive cooling system in traditional IR sensors.

C. Result

The *Melanophila acuminata* beetle detects infrared radiation from forest fires using specialized infrared sensory organs. This ability allows them to locate forest fires from distances up to 20 km.

Even more impressive, a single sensillum can detect a fire 10 km away with just 3.45×10^{-10} W of power—proof of its remarkable sensitivity and energy efficiency.

The study reveals the sensory organ structures of *Melanophila acuminata* beetle. Sensory pits contain 50–100 sensilla arranged in a highly organized pattern. Each sensillum consists of an

outer lipid layer forming a protective structure. A central tulip-shaped protein core, where IR absorption occurs. A polysaccharide base, where neuronal connections are present.

After these technique measurements, Gel electrophoresis revealed a single dominant protein of 15 kDa, later confirmed to be a fatty acid-binding protein. Raman spectroscopy and mass spectrometry confirmed the presence of aromatic amino acids (phenylalanine, tyrosine, tryptophan), which may contribute to infrared photon absorption. This unique infrared-sensitive protein was not found in any existing databases, suggesting it is a specialized evolutionary adaptation.

The beetle's infrared detection picks up signals at 3 μm and 10 μm , matching the infrared wavelengths from forest fires. And its infrared detection operates within the atmospheric transmission window (2.5–5 μm). The tulip-shaped protein core exhibited the highest absorption, confirming that proteins play a key role in IR detection.

The goal of the second paper's research is to develop an uncooled biomimetic IR sensor based on the beetle's sensory mechanism. The research presents the development of a biomimetic uncooled infrared (IR) sensor. However, the sensitivity estimations varied widely, ranging from 0.6 W/m² (behavioral experiments) to 150 W/m² (electrophysiological studies), creating significant uncertainty.

The developed sensor, using MEMS technology, achieved a modeled sensitivity of **$\sim 1 \text{ W/m}^2$** , which closely aligns with behavioral estimates, suggesting it realistically replicates the beetle's detection ability.

However, there are still many limitations, for example,

- The sensor is still in development and has not yet reached the sensitivity of state-of-the-art IR detectors.
- The compensation mechanism helps with temperature fluctuations, but background IR sources (e.g., the sun, hot surfaces) may still cause noise.
- The beetle may detect fires from 10–20 km, but the sensor is likely to have a shorter range due to technological constraints.

The biomimetic IR sensor, inspired by the Golay cell, was developed to mimic the infrared detection mechanism of beetles. Different cavity-filling fluids were tested, with hydrocarbons and gases showing better membrane deflection than water. Water had the lowest expansion and deflection ($\sim 0.04 \text{ nm}$), while hydrocarbons (n-pentane, toluene) increased deflection by 10x ($\sim 1.26 \text{ nm}$), and CO₂ performed best ($\sim 2.18 \text{ nm}$) due to its high thermal expansion. Theoretical calculations estimated a 350 Pa pressure increase in the water-filled cavity, with a maximum membrane deflection of $\sim 1 \text{ nm}$, similar to the beetle's receptor activation threshold. While the

sensor successfully replicates the photomechanic mechanism, it requires further optimization. The key challenge is that the tiny deflection (1–2 nm) demands highly sensitive read-out methods, and beetles' natural mechanosensitive neurons are still more efficient than any existing technology, making full biomimetic replication difficult.

D. Future Work & Challenges

Current applications

Biomimetic infrared (IR) sensors inspired by the *Melanophila acuminata* beetle have multiple real-world applications, including:

1. **Early wildfire detection:** These sensors can detect small temperature increases from fires before they spread, allowing for rapid response and fire prevention
 - They are more sensitive than traditional satellite systems and can detect heat at a much lower intensity compared to conventional sensors (as low as 1 W/m² compared to 2×10³ W/m² for current pyroelectric sensors)
 - The ability to detect fires from kilometers away enables real-time monitoring of remote forest areas
 - They can be integrated with AI powered fire prediction models to create automated alerts.
2. **Industrial fire safety:** Industries subject to fire hazards can use these sensors to detect overheating machinery or electrical faults
 - Unlike traditional temperature sensors that only detect surface heat, these sensors detect infrared radiation before flames appear.
 - They can be integrated into smart safety systems to trigger automatic shutdowns before fires start
3. **Military and surveillance:** There are also uses in military applications for detecting enemy heat signatures, missile launches, or thermal activity in the dark
 - The sensors don't emit signals, making them ideal for stealth surveillance.

4. **Environmental monitoring:** These sensors help scientists track climate change by detecting heat variations in ecosystems, volcanoes, and urban environments.

Future research objectives

- Miniaturisation and mass production: Development of nano-MEMS (Microelectromechanical Systems) technology will allow smaller, cheaper, and more widely available sensors
- AI powered fire prediction models: Combining IR sensor data with machine learning tools
- Integration with satellite and drone networks: This could enable global wildfire tracking in real time

Potential future applications

1. Smart Firefighting Technologies ; AI-driven smart firefighting systems could integrate these sensors to automatically analyze fire behavior and predict fire paths

Wearable IR detectors could assist firefighters by improving visibility in smoke-filled areas.

2. Smart Cities and Infrastructure

- Biomimetic IR sensors could be placed in public parks, power grids, and transportation systems to prevent fires caused by faulty wiring or overheating

3. Space Exploration and Defense

- a. Future Mars and lunar exploration missions could benefit from these sensors for detecting heat sources and preventing equipment malfunctions.

Challenges in the Field

1. Scalability and cost

- Current MEMS-based biomimetic sensors are expensive to produce in large quantities.
- Research is needed to simplify manufacturing while maintaining high sensitivity

2. Environmental factors

- Harsh environments (e.g dust, humidity, and extreme temperatures) can degrade sensor performance over time.
- Protective coatings and self-cleaning technologies are needed to improve long-term performance

3. Integration with Existing Technology

- Many firefighting and monitoring systems currently use traditional thermal cameras and satellite imaging, so integrating these novel sensors into current networks is a challenge

4. Interference from Background Heat Sources

- The IR compensation mechanism reduces false alarms, but more development is needed to distinguish wildfires from other heat sources such as the sun or industrial sites.

Future work of μ -biomimetic IR sensor:

1. Fusion bonding of two silicon wafers: the sensor design has 2 layers, one containing the pressure-sensitive chamber and another containing the capacitive sensor.

- Fusion bonding is needed to create a seamless integration of these two components. This ensures structural stability and consistent pressure transfer.

2. Bubble-free filling of the sensor

- The sensor contains liquid, which expands in response to IR radiation, creating a pressure-based detection system.
- Any trapped air bubbles could: reduce sensitivity or interfere with response time, making the sensor less reliable
- To solve this, precision microfluidic filling techniques need to be refined to totally eliminate air pockets.

3. Adapting PDMS Membranes into Fabrication

- PDMS (Polydimethylsiloxane) is a biocompatible polymer used in pressure-sensitive applications.
- It is a key component of the compensation mechanism, allowing slow liquid exchange to filter out background heat fluctuations.
- The challenge is finding a way to integrate this membrane into the current silicon-based MEMS fabrication process without affecting sensor performance.

E. Conclusion

This biomimetic case study highlights the biomimetic infrared (IR) sensory mechanism of the *Melanophila acuminata* beetle and its application in developing uncooled IR sensors. It is essential to notice that the biomimetic inspired sensor is effective and useful to solve some real world problems, including the early wildfire detection, military surveillance, and environmental monitoring.

The beetles can detect forest fires from up to 20 km using the specialized infrared sensory organs that efficiently convert the IR radiation in the change of mechanical pressure inside the liquid-filled microfluidic core. The subsequent membrane deformation is able to activate the mechanosensitive neurons, allowing fire detection through a bioelectric signal on the CNS.

The biomimetic method aims to overcome the limitations of conventional IR sensors, which rely on thermal detection and cooling mechanisms. Based on the biomimetic infrared (IR) sensory mechanism of the *Melanophila acuminata* beetle, MEMS-based uncooled IR sensor is developed to utilize the capacitive sensing for detecting the pressure-induced membrane deformations. Although the advantages and benefits of the biomimetic inspired solutions are noticeable and magnificent, the challenges, including achieving sensitivity comparable to natural beetle receptors, reducing interference from background IR sources, and improving sensor stability in real-world environments, behind this solution should not be neglected effortlessly.

F. Acknowledgement

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